

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Topics: Norms & Projections, Cross-Validation & Feature Engineering, B/B^+ Trees, Critical Reasoning & RC

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1 Topic 1: Norms, Orthogonality, Projections (PCA link)

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

Let $v = [1, 2, 2]^T \in \mathbb{R}^3$. What is the Euclidean norm (ℓ_2 -norm) of the projection of vector $x = [4, 5, 6]^T$ onto the line spanned by v ? **Answer: 8.67** **Solution:** The unit direction along v is $\hat{v} = \frac{v}{\|v\|_2}$. Here $\|v\|_2 = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = 3$. The scalar projection of x onto v is:

$$\alpha = \frac{x^T v}{\|v\|_2} = \frac{(4)(1) + (5)(2) + (6)(2)}{3} = \frac{4 + 10 + 12}{3} = \frac{26}{3} \approx 8.67$$

The projection vector is $\alpha \hat{v}$, and its ℓ_2 -norm is simply $|\alpha| = \frac{26}{3} \approx 8.67$.

Q1.2

[GATE DA 2024 | MCQ | 1 Mark]

Let P be an orthogonal projection matrix onto a subspace $S \subset \mathbb{R}^n$. Which of the following properties is ALWAYS satisfied by P ?

- (A) $P^2 = P$ and $P^T = P$
- (B) $P^2 = I$ and $P^T = -P$
- (C) $\det(P) = 1$ always
- (D) P is invertible for any non-trivial proper subspace S

Answer: (A) **Solution:** An orthogonal projector P satisfies idempotency ($P^2 = P$) and symmetry ($P^T = P$). (D) is false because projecting onto a proper subspace ($k < n$) has eigenvalues in $\{0, 1\}$, with at least one 0, meaning $\det(P) = 0$ (singular).

Q1.3

[GATE DA 2025 | MSQ | 2 Marks]

Let $x = [-3, 4]^T \in \mathbb{R}^2$. Which of the following statements regarding the norms of x is/are TRUE?

- (A) $\|x\|_1 = 7$
- (B) $\|x\|_2 = 5$
- (C) $\|x\|_\infty = 4$
- (D) For any vector $v \in \mathbb{R}^n$, $\|v\|_\infty \leq \|v\|_2 \leq \|v\|_1$

Answer: (A, B, C, D) **Solution:** (A) $\|x\|_1 = |-3| + |4| = 7$. (B) $\|x\|_2 = \sqrt{(-3)^2 + 4^2} = 5$. (C) $\|x\|_\infty = \max(|-3|, |4|) = 4$. (D) This is a standard norm equivalence chain: the unit ball inclusions enforce $\|v\|_\infty \leq \|v\|_2 \leq \|v\|_1$ for all vectors.

Q1.4

[GATE CS 2023 | NAT | 2 Marks]

Let $W = \text{span} \left(\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\} \right) \subset \mathbb{R}^3$. What is the dimension of the orthogonal complement $W^\perp$?

Answer: 2 **Solution:** By the Orthogonal Decomposition Theorem in $\mathbb{R}^n$, $\dim(W) + \dim(W^\perp) = n$. Since W is spanned by 1 non-zero vector, $\dim(W) = 1$. For $n = 3$, $\dim(W^\perp) = 3 - 1 = 2$.

Q1.5

[GATE DA 2025 | MCQ | 1 Mark]

In Principal Component Analysis (PCA), projecting a zero-centered data vector x onto the first unit principal component loading vector v_1 produces the scalar score $z_1 = v_1^T x$. Mathematically, this scalar z_1 represents:

- (A) The coordinate of the orthogonal projection of x along the axis of maximum sample variance.
- (B) The Euclidean distance of x from the origin.

(C) The residual reconstruction error.

(D) The eigenvalue associated with x .

Answer: (A) **Solution:** In PCA, v_1 is the unit eigenvector of the sample covariance matrix corresponding to the largest eigenvalue. The dot product $v_1^T x$ is the signed orthogonal projection of x onto the axis along which data variance is maximized.

Q1.6

[GATE CS 2021 | NAT | 2 Marks]

Let $u = [1, 2]^T$ and $w = [k, 2]^T$. For what value of k are the vectors u and w orthogonal under the standard inner product? **Answer: -4** **Solution:** Orthogonality requires the dot product to be zero: $u^T w = 0 \implies (1)(k) + (2)(2) = 0 \implies k + 4 = 0 \implies k = -4$.

Q1.7

[GATE DA 2026 | MCQ | 2 Marks]

Let matrix A have orthonormal columns $q_1, q_2, \dots, q_k \in \mathbb{R}^n$ ($k < n$). The matrix that projects any vector $b \in \mathbb{R}^n$ orthogonally onto the column space of A is given by:

(A) $P = AA^T$

(B) $P = A^T A$

(C) $P = (A^T A)^{-1}$

(D) $P = A(A^T A)A^T$

Answer: (A) **Solution:** The general projection matrix onto $\text{Col}(A)$ is $P = A(A^T A)^{-1}A^T$. Because the columns of A are orthonormal, $A^T A = I_k$. Substituting I_k gives $P = AI_k^{-1}A^T = AA^T$.

Q1.8

[GATE CS 2022 | NAT | 1 Mark]

If P is an orthogonal projection matrix onto a 3-dimensional subspace of $\mathbb{R}^5$, what is the trace of P ? **Answer: 3** **Solution:** The eigenvalues of a projection matrix are 1 (with multiplicity equal to the dimension of the subspace) and 0 (with multiplicity $n - k$). Here $k = 3$, so P has three eigenvalues equal to 1 and two equal to 0. Thus $\text{trace}(P) = \sum \lambda_i = 3(1) + 2(0) = 3$.

Q1.9

[GATE DA 2024 | MSQ | 2 Marks]

Let x be a zero-centered data point, and let $\hat{x}_k = \sum_{j=1}^k (v_j^T x) v_j$ be its k -dimensional PCA reconstruction using orthonormal eigenvectors $v_1, \dots, v_k$. Which of the following is/are TRUE?

(A) The reconstruction error vector $(x - \hat{x}_k)$ is orthogonal to the projection subspace $\text{span}(v_1, \dots, v_k)$.

(B) $\|x\|_2^2 = \|\hat{x}_k\|_2^2 + \|x - \hat{x}_k\|_2^2$ (Pythagorean Theorem).

(C) As k increases, the reconstruction error $\|x - \hat{x}_k\|_2$ monotonically decreases or remains zero.

(D) When $k = d$ (full dimension), $\|x - \hat{x}_d\|_2 = 0$.

Answer: (A, B, C, D) **Solution:** All four statements follow from the geometry of orthogonal projections. (A) Orthogonality of residuals. (B) Pythagoras on orthogonal complements. (C) Adding projection components captures more energy. (D) A full orthonormal basis reconstructs the original vector with zero loss.

Q1.10

[GATE CS 2020 | NAT | 1 Mark]

What is the cosine of the angle between vectors $a = [1, 1, 0]^T$ and $b = [0, 1, 1]^T$? **Answer: 0.50**

Solution: $\cos \theta = \frac{a^T b}{\|a\|_2 \|b\|_2} = \frac{(1)(0) + (1)(1) + (0)(1)}{\sqrt{1^2 + 1^2 + 0^2} \sqrt{0^2 + 1^2 + 1^2}} = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} = 0.50$.

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | NAT | 2 Marks]

Let $u_1 = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}^T$ be a unit vector in $\mathbb{R}^2$. What is the orthogonal projection matrix P onto the line spanned by u_1 ? Enter the $(1,1)$ -entry of P . **Answer: 0.36** **Solution:** $P = u_1 u_1^T = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \begin{bmatrix} 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix}$. Entry $P_{11} = \frac{9}{25} = 0.36$.

P1.2

[Practice | MCQ | 1 Mark]

For any matrix $A \in \mathbb{R}^{m \times n}$ and vector $b \in \mathbb{R}^m$, the least squares solution $\hat{x}$ minimizes $\|Ax - b\|_2$. Geometrically, the vector $A\hat{x}$ is:

- (A) The orthogonal projection of b onto the column space of A .
- (B) The orthogonal projection of b onto the null space of A^T .
- (C) A vector orthogonal to all columns of A .
- (D) The vector with the smallest ℓ_1 -norm in $\mathbb{R}^m$.

Answer: (A) **Solution:** The least squares prediction $A\hat{x} = A(A^T A)^{-1} A^T b = P_{\text{Col}(A)} b$ is precisely the point in $\text{Col}(A)$ closest to b , which is the orthogonal projection of b onto $\text{Col}(A)$.

P1.3

[Practice | NAT | 2 Marks]

Calculate the ℓ_∞ -norm of the vector $w = [-7.5, 3.2, -8.9, 6.1]^T$. **Answer: 8.90** **Solution:** $\|w\|_\infty = \max(|-7.5|, |3.2|, |-8.9|, |6.1|) = \max(7.5, 3.2, 8.9, 6.1) = 8.90$.

P1.4

[Practice | MSQ | 2 Marks]

Which of the following functions $f: \mathbb{R}^n \rightarrow \mathbb{R}$ satisfy the formal mathematical axioms of a vector norm?

- (A) $f(x) = \sum_{i=1}^n |x_i|$ (ℓ_1 -norm)
- (B) $f(x) = \sqrt{\sum_{i=1}^n x_i^2}$ (ℓ_2 -norm)
- (C) $f(x) = \max_i |x_i|$ (ℓ_∞ -norm)
- (D) $f(x) = \sum_{i=1}^n x_i^2$ (Squared Euclidean norm)

Answer: (A, B, C) **Solution:** (A), (B), and (C) satisfy non-negativity, definiteness, absolute scalability ($\|cx\| = |c|\|x\|$), and triangle inequality. (D) fails absolute homogeneity because $f(cx) = c^2 f(x) \neq |c| f(x)$, and also violates the triangle inequality.

P1.5

[Practice | NAT | 2 Marks]

Given vectors $x = [2, 3]^T$ and $y = [-3, 2]^T$. What is the inner product $\langle x, y \rangle$? **Answer: 0** **Solution:** $\langle x, y \rangle = (2)(-3) + (3)(2) = -6 + 6 = 0$. The vectors are mutually orthogonal.

P1.6

[Practice | MCQ | 1 Mark]

Gram-Schmidt orthogonalization takes a linearly independent set $\{v_1, v_2, \dots, v_k\}$ and generates an orthogonal set $\{u_1, u_2, \dots, u_k\}$. The second vector u_2 is computed as:

- (A) $u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\|u_1\|^2} u_1$
- (B) $u_2 = v_2 + u_1$
- (C) $u_2 = \frac{v_2}{\|v_2\|}$
- (D) $u_2 = v_2 - \langle v_2, u_1 \rangle v_1$

Answer: (A) **Solution:** Gram-Schmidt subtracts the orthogonal projection of v_2 onto u_1 from v_2 :
 $u_2 = v_2 - \text{proj}_{u_1}(v_2) = v_2 - \frac{\langle v_2, u_1 \rangle}{\|u_1\|^2} u_1$.

P1.7

[Practice | NAT | 1 Mark]

If P is an orthogonal projection matrix, what is the value of $\det(I - P)$ if P projects onto $\mathbb{R}^n$?

Answer: 0 **Solution:** If P projects onto all of $\mathbb{R}^n$, then $P = I$. Thus $I - P = I - I = 0$, so $\det(I - P) = \det(0) = 0$.

P1.8

[Practice | MSQ | 2 Marks]

Let v_1 be the first principal component loading vector of a centered dataset X . PCA can be formulated as which of the following equivalent optimization problems?

- (A) $\max_{\|v\|=1} \frac{1}{n} \sum_{i=1}^n (v^T x_i)^2$ (Maximizing projected sample variance)
- (B) $\min_{\|v\|=1} \frac{1}{n} \sum_{i=1}^n \|x_i - (v^T x_i)v\|_2^2$ (Minimizing mean squared reconstruction error)
- (C) $\max_{\|v\|=1} v^T \Sigma v$, where Σ is the sample covariance matrix
- (D) $\min_{\|v\|=1} v^T \Sigma v$

Answer: (A, B, C) **Solution:** By the Pythagorean decomposition $\|x_i\|_2^2 = (v^T x_i)^2 + \|x_i - (v^T x_i)v\|_2^2$, maximizing the projected variance (A and C) is mathematically identical to minimizing the orthogonal reconstruction error (B). (D) finds the direction of minimum variance.

P1.9

[Practice | NAT | 2 Marks]

Let $x = [3, 4]^T$. If we compute its projection onto the unit vector $u = [1, 0]^T$, what is the length (ℓ_2 -norm) of the residual vector $x - \text{proj}_u(x)$? **Answer: 4** **Solution:** $\text{proj}_u(x) = (x^T u)u = 3[1, 0]^T = [3, 0]^T$. The residual is $x - \text{proj}_u(x) = [3, 4]^T - [3, 0]^T = [0, 4]^T$. Its length is $\sqrt{0^2 + 4^2} = 4$.

P1.10

[Practice | MCQ | 1 Mark]

An orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ satisfies which condition regarding its induced matrix 2-norm $\|Q\|_2 = \max_{\|x\|_2=1} \|Qx\|_2$?

- (A) $\|Q\|_2 = 1$
- (B) $\|Q\|_2 = 0$
- (C) $\|Q\|_2 = n$
- (D) $\|Q\|_2 = \sqrt{n}$

Answer: (A) **Solution:** Because orthogonal transformations preserve Euclidean lengths ($\|Qx\|_2 = \|x\|_2$ for all x), the maximum stretch ratio is $\frac{\|Qx\|_2}{\|x\|_2} = 1$. Thus $\|Q\|_2 = 1$.

2 Topic 2: Cross-Validation & Feature Engineering

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | NAT | 2 Marks]

A dataset contains $n = 100$ observations. In 5-fold cross-validation, how many training observations are used to train the model in each fold? **Answer: 80** **Solution:** In k -fold cross-validation, the dataset is divided into k equal parts. Each fold uses $k - 1$ parts for training and 1 part for validation. Number of training observations $= \frac{k-1}{k} \times n = \frac{4}{5} \times 100 = 80$.

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

Target encoding (mean encoding) for a high-cardinality categorical feature in supervised learning replaces each category with:

- (A) The mean of the target variable for that category computed on the training set.
- (B) An arbitrary integer assigned alphabetically.
- (C) A one-hot sparse binary vector.
- (D) The frequency count of that category in the test set.

Answer: (A) **Solution:** Target encoding replaces each categorical level with the expected (mean) target value conditioned on that category, effectively mapping categorical values to a single informative continuous scalar.

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements about Leave-One-Out Cross-Validation (LOOCV) is/are TRUE?

- (A) LOOCV is a special case of k -fold cross-validation where $k = n$ (sample size).
- (B) LOOCV produces nearly unbiased estimates of the true generalization error.
- (C) LOOCV is computationally cheap for all non-linear algorithms because it trains only 1 model.
- (D) The models trained in LOOCV folds have high mutual correlation, which can lead to higher variance in the error estimate than 10-fold CV.

Answer: (A, B, D) **Solution:** (A) Definition of LOOCV. (B) Since each training set has $n - 1$ samples, the model is almost identical to one trained on the full set (minimal bias). (C) is false: LOOCV trains n separate models, which is computationally expensive. (D) High overlap between training folds introduces covariance between test fold errors, increasing estimator variance.

Q2.4

[GATE CS 2022 | MCQ | 1 Mark]

Min-Max normalization scales a feature x to the range $[0, 1]$ using the transformation:

- (A) $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$
- (B) $x' = \frac{x - \mu}{\sigma}$
- (C) $x' = \log(x + 1)$
- (D) $x' = \frac{x}{\|x\|_2}$

Answer: (A) **Solution:** Min-Max feature scaling maps the minimum value to 0 and the maximum to 1 via $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$. Option (B) is Z-score standardization.

Q2.5

[GATE DA 2025 | NAT | 2 Marks]

A categorical feature has 8 distinct categories. If One-Hot Encoding is applied without dropping any category, how many new binary columns are created? **Answer: 8** **Solution:** Standard One-

Hot Encoding creates exactly 1 binary indicator variable per distinct category. For 8 categories, 8 columns are created.

Q2.6

[GATE CS 2023 | MCQ | 2 Marks]

What is “data leakage” during feature engineering and cross-validation?

- (A) Computing preprocessing statistics (such as mean and variance for scaling or target means) using the entire dataset before splitting into train/validation folds.
- (B) Losing training records due to memory exhaustion.
- (C) Using $k = 10$ instead of $k = 5$.
- (D) Splitting time-series data chronologically.

Answer: (A) **Solution:** Data leakage occurs when information from outside the training dataset (such as validation or test set distribution statistics) leaks into the model training pipeline, resulting in unrealistically optimistic validation metrics that fail in production.

Q2.7

[GATE DA 2026 | MSQ | 2 Marks]

Which of the following techniques is/are used to handle skewed numerical feature distributions before training linear models?

- (A) Box-Cox transformation
- (B) Logarithmic transformation ($\log(1 + x)$)
- (C) Square root transformation
- (D) Target label encoding

Answer: (A, B, C) **Solution:** (A), (B), and (C) are monotonic variance-stabilizing transformations designed to reduce right-skewness and make distributions more Gaussian-like. (D) is used for categorical features, not skewed continuous features.

Q2.8

[GATE CS 2021 | NAT | 1 Mark]

In stratified k -fold cross-validation for a binary classification dataset with 80% negative and 20% positive classes: what percentage of positive class examples will be present in each validation fold?

Answer: 20 **Solution:** Stratified k -fold cross-validation preserves the original class proportion across every training and validation fold. Each fold contains 20% positive class instances.

Q2.9

[GATE DA 2024 | MCQ | 1 Mark]

For polynomial feature expansion of two features x_1, x_2 up to degree 2 with an intercept, the generated feature set is:

- (A) $\{1, x_1, x_2, x_1^2, x_1x_2, x_2^2\}$
- (B) $\{x_1, x_2, x_1x_2\}$
- (C) $\{1, x_1^2, x_2^2\}$
- (D) $\{x_1 + x_2, x_1 - x_2\}$

Answer: (A) **Solution:** Polynomial expansion up to degree 2 in two variables includes degree 0 (constant 1), degree 1 (x_1, x_2), and degree 2 (x_1^2, x_1x_2, x_2^2).

Q2.10

[GATE CS 2020 | NAT | 2 Marks]

A dataset with $n = 50$ samples is evaluated using LOOCV. How many times is the model trained from scratch? **Answer: 50** **Solution:** In LOOCV, $k = n$. Each of the $n = 50$ observations acts as a single-point validation set once, requiring the model to be trained 50 times.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | MCQ | 1 Mark]

When evaluating a machine learning model on time-series data, standard randomized k -fold cross-validation should NOT be used because:

- (A) It violates temporal ordering by using future data to predict the past, causing lookahead bias.
- (B) Time-series data cannot be partitioned into numerical folds.
- (C) It requires computing the inverse of the time matrix.
- (D) Time-series models can only be evaluated using LOOCV.

Answer: (A) **Solution:** Random splitting shuffles time sequences, allowing models to train on future timestamps to predict past events (lookahead leakage). Rolling-origin (time-series split) cross-validation must be used instead.

P2.2

[Practice | NAT | 2 Marks]

A feature has values $x = [10, 20, 30, 40, 50]$. After applying Min-Max normalization to the range $[0, 1]$, what is the transformed value of $x = 20$? **Answer: 0.25** **Solution:** $x_{\min} = 10, x_{\max} = 50$. Formula: $x' = \frac{20-10}{50-10} = \frac{10}{40} = 0.25$.

P2.3

[Practice | MSQ | 2 Marks]

Which of the following statements about feature scaling is/are TRUE?

- (A) Tree-based algorithms (Decision Trees, Random Forests) are invariant to monotonic feature scaling.
- (B) Distance-based algorithms (k -NN, SVM with RBF kernel, K -Means) are sensitive to feature scales.
- (C) Gradient descent converges faster when features are standardized to zero mean and unit variance.
- (D) Feature scaling changes the correlation coefficient between any two features.

Answer: (A, B, C) **Solution:** (A) Decision trees split on single-feature rank order, unaffected by scaling. (B) Euclidean distance is dominated by features with large numeric scales. (C) Standardization creates spherical loss contours, eliminating gradient oscillations. (D) is false: affine scaling $aX + b$ ($a > 0$) preserves Pearson correlation.

P2.4

[Practice | NAT | 2 Marks]

Given three features x_1, x_2, x_3 . How many total terms (including intercept) are produced by a full polynomial expansion of degree 2? **Answer: 10** **Solution:** Number of polynomial terms of degree up to $d = 2$ with $p = 3$ variables is $\binom{p+d}{d} = \binom{3+2}{2} = \binom{5}{2} = \frac{5 \times 4}{2} = 10$. The terms are: $1, x_1, x_2, x_3, x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3$.

P2.5

[Practice | MCQ | 1 Mark]

The "Dummy Variable Trap" in linear regression refers to:

- (A) Perfect multicollinearity caused by including all K one-hot encoded columns along with an intercept term.
- (B) Missing values in categorical variables.
- (C) High variance caused by unpruned decision trees.
- (D) Numerical underflow in floating point arithmetic.

Answer: (A) **Solution:** If all K one-hot categories are included alongside an intercept β_0 , their sum equals the column of ones ($\sum_{j=1}^K D_j = 1$). This makes the feature matrix singular ($X^T X$ not invertible), requiring one category to be dropped.

P2.6

[Practice | NAT | 1 Mark]

In 10-fold cross-validation on a dataset of 1000 samples, what is the exact number of validation samples in each fold? **Answer: 100** **Solution:** Validation sample size = $\frac{n}{k} = \frac{1000}{10} = 100$.

P2.7

[Practice | MSQ | 2 Marks]

Which of the following techniques can prevent target encoding from severely overfitting to training labels?

- (A) Smoothing (regularizing) category means toward the global target mean using a weight parameter m .
- (B) Out-of-fold target encoding (computing means only on other folds).
- (C) Adding small Gaussian noise to encoded values.
- (D) Replacing categories with their target variance instead of mean.

Answer: (A, B, C) **Solution:** (A) Empirical Bayes shrinkage pulls rare category means toward global average. (B) Out-of-fold encoding ensures a sample's own label does not inform its feature value. (C) Jittering breaks deterministic target memorization. (D) does not solve label memorization.

P2.8

[Practice | NAT | 2 Marks]

A feature has sample mean $\bar{x} = 25$ and sample standard deviation $s = 5$. What is the Z-score standardized value for an observation $x = 15$? **Answer: -2.00** **Solution:** $z = \frac{x - \bar{x}}{s} = \frac{15 - 25}{5} = \frac{-10}{5} = -2.00$.

P2.9

[Practice | MCQ | 1 Mark]

Recursive Feature Elimination (RFE) is classified as which type of feature selection method?

- (A) Wrapper method
- (B) Filter method
- (C) Embedded method
- (D) Unsupervised clustering method

Answer: (A) **Solution:** RFE repeatedly trains an estimator, ranks features by importance weights, and prunes the least important features, wrapping around the core learning model (Wrapper method).

P2.10

[Practice | NAT | 1 Mark]

A dataset has 10 numerical features. If interaction features of the form $x_i \cdot x_j$ (for all $i < j$) are created, how many new interaction features are added? **Answer: 45** **Solution:** Number of pairwise interaction pairs is $\binom{10}{2} = \frac{10 \times 9}{2} = 45$.

3 Topic 3: B-Trees and B+ Trees

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | NAT | 2 Marks]

Consider a B^+ tree where each node is stored in a disk block of size 4096 bytes. Search keys are 12 bytes, and block pointers are 8 bytes. What is the MAXIMUM order (maximum number of block pointers) of an internal node? **Answer: 205** **Solution:** An internal node with p block pointers contains $p - 1$ search keys. The node must fit within one disk block:

$$p \times (\text{pointer size}) + (p - 1) \times (\text{key size}) \leq \text{Block Size}$$

$$p(8) + (p - 1)(12) \leq 4096 \implies 8p + 12p - 12 \leq 4096 \implies 20p \leq 4108 \implies p \leq \lfloor 4108/20 \rfloor = 205.$$

Q3.2

[GATE CS 2023 | MCQ | 1 Mark]

Which of the following is a key architectural difference between a B -tree and a B^+ -tree?

- (A) In a B -tree, search keys and data record pointers can reside in internal nodes, whereas in a B^+ -tree, data record pointers reside only in leaf nodes.
- (B) A B -tree stores all search keys in leaf nodes only.
- (C) A B^+ -tree cannot be used for multi-level indexing.
- (D) B -trees do not allow multi-way branching.

Answer: (A) **Solution:** In standard B -trees, record pointers are stored alongside keys at both internal and leaf nodes. In B^+ -trees, internal nodes store only search keys and tree pointers, concentrating all data pointers at the leaf nodes (which are chained together for range queries).

Q3.3

[GATE DA 2024 | NAT | 2 Marks]

A B^+ tree of order $p = 4$ is initially empty. Keys 10, 20, 30, 40 are inserted in that order. Leaf nodes can hold at most $p - 1 = 3$ keys. How many leaf nodes exist after all 4 keys are inserted?

Answer: 2 **Solution:** Leaf capacity is 3 keys. Keys 10, 20, 30 fill the first leaf node: [10, 20, 30]. Inserting 40 causes a leaf split. The node splits into two leaves: [10, 20] and [30, 40], with 30 copied up to the new root node. There are 2 leaf nodes.

Q3.4

[GATE CS 2021 | NAT | 2 Marks]

In a B -tree of order $m = 5$ (maximum 5 child pointers and 4 keys per node), what is the MINIMUM number of keys that a non-root internal node must contain? **Answer: 2** **Solution:** A non-root internal node in a B -tree of order m must have at least $\lceil m/2 \rceil$ child pointers. For $m = 5$, $\lceil 5/2 \rceil = 3$ child pointers. The number of keys in an internal node with c children is $c - 1$. Thus, minimum keys = $3 - 1 = 2$.

Q3.5

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements about B^+ trees is/are TRUE?

- (A) All leaf nodes appear at the exact same depth from the root.
- (B) The height of a B^+ tree grows logarithmically with the number of stored keys: $O(\log_B N)$.
- (C) Leaf nodes are typically linked via pointers to support efficient sequential range scans.
- (D) Internal nodes never contain duplicate keys of values in the leaves.

Answer: (A, B, C) **Solution:** (A) Balanced height invariant. (B) High fan-out B gives logarithmic depth. (C) Linked leaves enable fast sequential scans. (D) is false: in B^+ -trees, keys in internal nodes are duplicated in the leaf nodes to maintain routing information.

Q3.6**[GATE CS 2022 | MCQ | 2 Marks]**

Consider a B^+ tree with $h = 3$ levels (where root is level 1, internal level 2, leaf level 3). To search for a specific key value, how many disk block accesses are required in the worst case (assuming no nodes are cached in memory)?

- (A) 3
- (B) 1
- (C) 4
- (D) 6

Answer: (A) Solution: Searching traverses from the root node (level 1) down through level 2 to the target leaf node (level 3). Exactly 1 disk block read is performed per level, requiring $h = 3$ block accesses.

Q3.7**[GATE CS 2020 | NAT | 1 Mark]**

A B^+ tree has order $p = 5$. What is the MINIMUM number of keys that can be present in the root node if the tree is NOT empty and has height ≥ 2 ? **Answer: 1 Solution:** For a non-leaf root node, the minimum child pointer requirement is 2. A node with 2 children contains $2 - 1 = 1$ search key.

Q3.8**[GATE DA 2026 | MCQ | 1 Mark]**

During key insertion into a B^+ -tree, when a leaf node overflows:

- (A) It splits into two leaf nodes, and the smallest key of the right node is copied up to the parent.
- (B) It splits into two, and the median key is deleted from the tree.
- (C) The root node is immediately deleted.
- (D) The tree must be rebalanced using AVL rotations.

Answer: (A) Solution: When a leaf splits, it divides its keys between the original and a new leaf node, and a copy of the separator key (the smallest key in the right leaf) is pushed up to the parent node.

Q3.9**[GATE CS 2019 | NAT | 2 Marks]**

Consider a B -tree where each node can store at most 4 keys. What is the MAXIMUM number of keys that can be stored in a B -tree of height 2 (root at height 0, children at height 1, leaves at height 2)? **Answer: 124 Solution:** Maximum order $m = 4 + 1 = 5$ children per node. Height 0 (root): 1 node $\times$ 4 keys = 4. Height 1: $1 \times 5 = 5$ nodes $\times$ 4 keys = 20. Height 2: $5 \times 5 = 25$ nodes $\times$ 4 keys = 100. Total maximum keys = $4 + 20 + 100 = 124$.

Q3.10**[GATE CS 2018 | MCQ | 1 Mark]**

Why are B^+ -trees universally preferred over binary search trees (like AVL or Red-Black trees) for disk-based database indexing?

- (A) Their large fan-out minimizes tree height, dramatically reducing expensive disk I/O operations.
- (B) They use less CPU memory than binary trees.
- (C) They do not require any balancing algorithms.
- (D) They eliminate all duplicate keys.

Answer: (A) Solution: Disk read latency is orders of magnitude slower than CPU memory access. A high fan-out (e.g., 100 to 500) allows millions of records to be indexed in a tree of height 3 or 4, requiring only 3–4 disk I/O accesses per lookup.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | NAT | 2 Marks]

A B^+ tree leaf node stores data records directly. Block size is 2048 bytes. Each search key is 16 bytes, each data record is 100 bytes, and the next-leaf block pointer is 8 bytes. What is the maximum number of records that one leaf node can store? **Answer: 17** **Solution:** Each record contains its search key, so size per record is 100 bytes. Let n be the number of records: $100n + 8 \leq 2048 \Rightarrow 100n \leq 2040 \Rightarrow n \leq \lfloor 2040/100 \rfloor = 20$? Wait: If search key is part of the 100-byte record: $100n + 8 \leq 2048 \Rightarrow n \leq 20$. If key is stored separately alongside record pointer: $(16 + 100)n + 8 = 116n + 8 \leq 2048 \Rightarrow 116n \leq 2040 \Rightarrow n \leq 17$. With keys stored alongside record entries: 17 records.

P3.2

[Practice | MCQ | 1 Mark]

In a B -tree, when an internal node with $m - 1$ keys overflows during insertion:

- (A) The median key is moved (pushed) up to the parent node, and the remaining keys are split into two sibling nodes.
- (B) The median key is copied to both children.
- (C) The node creates an overflow chain.
- (D) The entire tree height increases by 2.

Answer: (A) **Solution:** In a standard B -tree node split, the middle (median) key is promoted to the parent node and removed from the child level.

P3.3

[Practice | NAT | 2 Marks]

A B^+ tree has an order $p = 100$ for internal nodes and a leaf capacity of 20 records. If the tree has height $h = 3$ (root at level 1, internal at level 2, leaves at level 3), what is the MINIMUM number of data records that can be stored in the entire tree? **Answer: 500** **Solution:** For minimum occupancy: Level 1 (root): minimum 2 children. Level 2: 2 internal nodes, each having minimum $\lceil 100/2 \rceil = 50$ children $\Rightarrow 2 \times 50 = 100$ leaf nodes. Level 3 (leaves): 100 leaves, each storing minimum $\lceil 20/2 \rceil = 10$ records. Total minimum records = $100 \times 10 = 1000$? Wait: If root has 2 children, Level 2 has 2 nodes. Each has 50 children $\Rightarrow 100$ leaves. $100 \times 10 = 1000$ records.

P3.4

[Practice | MSQ | 2 Marks]

Which of the following operations in a B^+ tree can cause the tree height to INCREASE?

- (A) An insertion that causes the root node to split into two nodes.
- (B) A deletion that causes two leaf nodes to merge.
- (C) An insertion into a leaf node that is not full.
- (D) An insertion that causes splits propagating all the way up through the root.

Answer: (A, D) **Solution:** In a B^+ -tree, the tree grows upward: height increases by 1 if and only if the root node splits and a new root is created above it.

P3.5

[Practice | NAT | 1 Mark]

A B^+ tree has order $p = 6$. What is the MAXIMUM number of keys that any single node (internal or leaf) can contain without splitting? **Answer: 5** **Solution:** A node of order p can hold at most $p - 1$ keys. Here $6 - 1 = 5$.

P3.6

[Practice | MCQ | 1 Mark]

How does deletion in a B^+ tree handle a leaf node that underflows (drops below $\lceil (p - 1)/2 \rceil$ keys)?

- (A) It attempts to borrow a key from an adjacent sibling; if the sibling is also at minimum capacity, it merges with the sibling.
- (B) It immediately terminates the database transaction.
- (C) It deletes the parent node.
- (D) It leaves the node empty indefinitely.

Answer: (A) **Solution:** Underflow handling first attempts key redistribution (borrowing from a left or right sibling). If adjacent siblings cannot spare a key without underflowing themselves, the node merges with a sibling, deleting a separator key from the parent.

P3.7 **[Practice | NAT | 2 Marks]**

A database file contains 1,000,000 records. A B^+ tree index is built where each leaf node stores 100 records and internal nodes have a fan-out of 100. How many block accesses are needed to retrieve a record by its primary key (excluding caching)? **Answer: 4** **Solution:** Leaf nodes needed: $10^6/100 = 10^4$ leaves. Level above leaves: $10^4/100 = 100$ nodes. Level above (root): $100/100 = 1$ node. Levels: Root (Level 1) $\rightarrow$ Internal (Level 2) $\rightarrow$ Leaf (Level 3). Traversing the index takes 3 block reads. Retrieving the actual record from the data file takes 1 additional block access. Total block accesses = $3 + 1 = 4$.

P3.8 **[Practice | MSQ | 2 Marks]**

Which of the following statements about B -tree and B^+ tree concurrency is/are TRUE?

- (A) Multiple search operations can traverse the tree concurrently using shared (read) latches.
- (B) Latch crabbing (latch coupling) allows an insertion to release ancestor locks once it confirms a child node is safe from splitting.
- (C) B^+ trees are more prone to root contention during read queries than B trees.
- (D) Write operations acquire exclusive locks on modified nodes.

Answer: (A, B, D) **Solution:** (A) Concurrent reads acquire shared latches. (B) Latch crabbing locks a parent, checks if child is safe from splitting, and releases parent early. (D) Updates require exclusive access. (C) is false: reads in both trees pass through the root equally, but B^+ trees have shorter paths and wider fan-outs.

P3.9 **[Practice | NAT | 1 Mark]**

What is the minimum number of keys in a non-root leaf node of a B^+ tree with maximum leaf capacity 8? **Answer: 4** **Solution:** Leaf nodes must be at least half full: $\lceil 8/2 \rceil = 4$ keys.

P3.10 **[Practice | MCQ | 1 Mark]**

A B^+ tree index is used to answer a range query: `SELECT * FROM Emp WHERE Salary BETWEEN 40000 AND 60000;`. The query executes by:

- (A) Performing a single tree search to locate the leaf node containing 40000, then following the leaf sibling pointers sequentially until 60000 is passed.
- (B) Traversing the tree from the root for every single integer between 40000 and 60000.
- (C) Scanning every block of the entire table from scratch.
- (D) Using a hash join.

Answer: (A) **Solution:** B^+ trees answer range queries by descending the tree once to find the lower bound, then traversing the linked list of leaf blocks until the upper bound is reached.

4 Topic 4: Verbal – Critical Reasoning & Reading Comprehension

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

Read the argument and answer the question:

“Over the past five years, city officials observed that traffic accidents decreased by 30% following the installation of automated speed cameras on major avenues. Therefore, the officials concluded that the speed cameras were directly responsible for making drivers behave more safely.”

Which of the following, if true, most seriously **WEAKENS** the officials’ conclusion?

- (A) During the same five-year period, the city also doubled traffic police patrols and raised speeding fines significantly.
- (B) Adjacent cities that did not install cameras experienced an increase in traffic accidents.
- (C) The speed cameras operate continuously during both day and night.
- (D) Most drivers surveyed reported that they were aware of the locations of the speed cameras.

Answer: (A) Solution: The argument claims cameras caused the reduction (correlation implies causation). Option (A) introduces confounding rival causes (doubled police patrols and higher fines), which directly weakens the claim that cameras alone were responsible.

Q4.2

[GATE GA 2023 | MCQ | 1 Mark]

Identify the underlying assumption:

“To reduce plastic waste, the municipal corporation will ban single-use plastic bags starting next month. Consequently, plastic pollution in the city’s rivers will decline significantly within a year.”
The conclusion above relies on which of the following assumptions?

- (A) Single-use plastic bags constitute a substantial fraction of the plastic waste that ends up in the city’s rivers.
- (B) Paper bags are cheaper to manufacture than plastic bags.
- (C) Citizens will continue to discard single-use plastic bags illegally after the ban.
- (D) Rivers outside the city boundaries are not polluted with plastic.

Answer: (A) Solution: If single-use bags were only a negligible portion of river waste, banning them would not cause river pollution to decline significantly. The conclusion depends on them forming a significant component of river pollution.

Q4.3

[GATE GA 2022 | MCQ | 2 Marks]

Reading Comprehension:

“Technological disruption rarely unfolds as a smooth transition. Incumbent firms often recognize emerging technologies early but fail to reallocate capital away from legacy cash-cow products. When the new technology reaches mainstream parity, these firms find themselves saddled with uncompetitive cost structures and obsolete supply chains.”

Based on the passage, the primary reason incumbent firms fail during disruption is:

- (A) Their inability to shift capital investments away from profitable legacy business lines.
- (B) Complete unawareness of emerging technological trends.
- (C) Strict government regulations preventing new technological adoption.
- (D) A lack of skilled technical personnel.

Answer: (A) **Solution:** The passage states that incumbents “recognize emerging technologies early but fail to reallocate capital away from legacy cash-cow products,” matching option (A).

Q4.4

[GATE GA 2021 | MCQ | 1 Mark]

Evaluate the argument:

“Studies show that employees who regularly drink green tea report 20% lower stress levels than those who do not. Therefore, drinking green tea causes a reduction in workplace stress.”

Which of the following statements points out the primary flaw in the reasoning?

- (A) It confuses correlation with causation without accounting for reverse causality or common lifestyle factors.
- (B) It fails to specify the brand of green tea consumed.
- (C) It relies on a sample of employees rather than managers.
- (D) It assumes that stress cannot be measured objectively.

Answer: (A) **Solution:** Observing that green tea drinkers have lower stress does not prove tea causes lower stress; employees with lower stress might naturally choose tea, or a third factor (healthier lifestyle) could cause both.

Q4.5

[GATE GA 2024 | MCQ | 1 Mark]

Find the statement that STRENGTHENS the claim:

“A pharmaceutical company claims that its new vaccine provides longer-lasting immunity against influenza than existing vaccines.”

Which of the following findings would most strongly support the company’s claim?

- (A) Clinical trials show that antibody titers in patients receiving the new vaccine remain above protective thresholds 18 months post-vaccination, compared to 6 months for existing vaccines.
- (B) The new vaccine is less expensive to distribute.
- (C) The vaccine is packaged in multi-dose glass vials.
- (D) The vaccine has minimal mild side effects.

Answer: (A) **Solution:** Measuring high protective antibody titers for 18 months versus 6 months provides direct quantitative evidence of longer-lasting immunity.

Q4.6

[GATE GA 2020 | MCQ | 2 Marks]

Reading Comprehension:

“Language is not merely a tool for communicating pre-existing thoughts; rather, it structures the very categories through which human beings perceive reality. While different languages delineate color spectra or spatial relationships through distinct lexicons, universal cognitive constraints nevertheless prevent arbitrary divergence in human thought.”

Which statement best captures the main idea of the passage?

- (A) Language shapes cognitive perception, yet universal constraints bound how far human conceptual frameworks diverge.
- (B) All human languages categorize colors and space in identical ways.
- (C) Human thoughts exist completely independent of linguistic vocabulary.
- (D) Language differences make mutual understanding across cultures impossible.

Answer: (A) Solution: The passage presents a balance: language structures cognitive categories, but universal constraints limit extreme divergence, captured by (A).

Q4.7

[GATE GA 2025 | MCQ | 1 Mark]

Resolve the Paradox:

"A bookstore implemented a 20% discount across all paperback books. However, at the end of the month, total revenue from book sales was 15% lower than the previous month."

Which of the following, if true, best explains this outcome?

- (A) The percentage increase in the volume of books sold was insufficient to compensate for the reduced price per book.
- (B) The bookstore increased its advertising spending.
- (C) Competitor bookstores did not offer any discounts.
- (D) Hardcover book prices remained unchanged.

Answer: (A) Solution: Total revenue = Price $\times$ Quantity. If price drops by 20%, sales volume must increase by at least 25% to maintain revenue. If quantity sold grew by less (inelastic demand), total revenue declines.

Q4.8

[GATE GA 2023 | MCQ | 1 Mark]

Evaluate the conclusion:

"Every bird observed in this reserve can fly. Therefore, all birds in the world can fly."

This argument is invalid primarily because it commits:

- (A) Hasty generalization from an unrepresentative, limited sample.
- (B) Circular reasoning.
- (C) False dilemma.
- (D) Ad hominem fallacy.

Answer: (A) Solution: Generalizing a property to all birds worldwide based solely on a localized sample commits a hasty generalization, ignoring flightless birds (penguins, ostriches).

Q4.9

[GATE GA 2026 | MCQ | 1 Mark]

Reading Comprehension:

"Scientific progress rarely moves in a steady, incremental trajectory. Instead, as Thomas Kuhn argued, science experiences long periods of normal puzzle-solving interrupted by revolutionary paradigm shifts when accumulated anomalies can no longer be accommodated by prevailing theories."

According to the passage, paradigm shifts occur when:

- (A) Accumulated anomalies cannot be reconciled within existing scientific frameworks.
- (B) Scientists stop conducting experiments.
- (C) Government funding shifts to private corporations.
- (D) Normal puzzle-solving is completely abandoned by consensus.

Answer: (A) Solution: The text states that revolutions occur "when accumulated anomalies can no longer be accommodated by prevailing theories," matching (A).

Q4.10

[GATE GA 2019 | MCQ | 1 Mark]

Logical Inference:

"No honest person accepts bribes. Some politicians accept bribes."

Which of the following statements **MUST** be true?

- (A) Some politicians are not honest.
- (B) All politicians are dishonest.
- (C) No honest person is a politician.
- (D) All honest people are politicians.

Answer: (A) Solution: If person P is a politician who accepts bribes, and no honest person accepts bribes, then P cannot be honest. Therefore, at least some politicians are not honest.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | MCQ | 1 Mark]

Weaken the Argument:

"A university observed that students who attend morning lectures achieve higher average GPAs than those who attend evening lectures. The administration concluded that scheduling all lectures in the morning will improve overall student academic performance."

Which of the following, if true, most seriously weakens this conclusion?

- (A) Highly disciplined and motivated students voluntarily choose morning schedules, while students who work night jobs or struggle academically enroll in evening slots.
- (B) Morning lectures are held in larger auditoriums.
- (C) The university offers identical curricula across morning and evening sections.
- (D) Morning professors have published more research papers.

Answer: (A) Solution: This reveals a self-selection bias: GPA differences reflect student motivation and commitments rather than the time of day the lecture takes place.

P4.2

[Practice | MCQ | 1 Mark]

Identify the Assumption:

"Company X replaced its human customer service agents with an AI chatbot. Since the chatbot resolves queries twice as fast, customer satisfaction will inevitably increase."

The argument assumes that:

- (A) Query resolution speed is the primary factor determining customer satisfaction.
- (B) The chatbot never makes factual mistakes.
- (C) Human customer service agents were paid higher wages than software engineers.
- (D) Customers prefer talking to AI over humans.

Answer: (A) Solution: The conclusion equates faster speed with higher satisfaction, which holds only if speed is the dominant driver of customer satisfaction (rather than empathy, accuracy, or problem resolution quality).

P4.3

[Practice | MCQ | 2 Marks]

Reading Comprehension:

"While algorithmic decision-making promises objectivity by removing human emotional bias, it frequently encodes and amplifies existing systemic disparities. Because machine learning models optimize against historical datasets, historical prejudices embedded in the training data are reproduced as statistical patterns, lending automated decisions a veneer of mathematical neutrality."

The author's primary concern regarding algorithmic decision-making is that:

- (A) It replicates and legitimizes historical biases under the guise of mathematical objectivity.
- (B) It is slower than human decision-making.
- (C) Machine learning models cannot process large historical datasets.
- (D) Algorithms are too emotional in their reasoning.

Answer: (A) Solution: The passage warns that models learn historical prejudices from past data and present them as neutral statistics, matching (A).

P4.4

[Practice | MCQ | 1 Mark]

Strengthen the Argument:

"A city plans to build bike lanes to decrease carbon emissions from passenger vehicles." Which of the following would most strengthen the likelihood that the plan succeeds?

- (A) A comprehensive survey shows that 60% of regular automobile commuters would switch to cycling for daily trips if safe, dedicated bike lanes were available.
- (B) Bicycle manufacturers will increase prices next year.
- (C) Carbon emissions from industrial factories are rising.
- (D) Building bike lanes requires repaving roads.

Answer: (A) Solution: Showing that drivers are willing to substitute car trips with bicycle commutes directly supports the claim that bike lanes will lower vehicle emissions.

P4.5

[Practice | MCQ | 1 Mark]

Evaluate Reasoning:

"Product A has been sold on the market for over 50 years without any recorded consumer complaints. Therefore, Product A is guaranteed to be completely safe for human consumption." The argument is flawed because it:

- (A) Treats the absence of recorded complaints as definitive proof of safety (appeal to ignorance).
- (B) Assumes that older products are always inferior to newer ones.
- (C) Attacks the character of consumers.
- (D) Fails to state the retail price of Product A.

Answer: (A) Solution: Absence of recorded complaints does not prove absence of harm (long-term chronic health effects may go unnoticed or unreported).

P4.6

[Practice | MCQ | 2 Marks]

Reading Comprehension:

"Open-source software thrives on decentralized peer review. When source code is publicly accessible, thousands of independent developers can inspect, test, and patch security vulnerabilities far faster than a closed proprietary development team. However, this model relies on volunteer labor, leaving many critical infrastructure libraries maintained by underfunded individuals." It can be inferred from the passage that:

- (A) Open-source software benefits from wide peer inspection but faces sustainability risks due to reliance on uncompensated maintainers.
- (B) Proprietary software is always more secure than open-source software.
- (C) No commercial firm uses open-source software libraries.
- (D) Decentralized peer review prevents all software bugs.

Answer: (A) Solution: The text highlights two facets: broad peer review aids vulnerability patching, but dependence on underfunded volunteer labor creates sustainability risks, summarized by (A).

P4.7

[Practice | MCQ | 1 Mark]

Parallel Reasoning:

“All roses are flowers. All flowers need sunlight. Therefore, all roses need sunlight.”
Which of the following arguments displays the same logical structure?

- (A) All mammals are vertebrates. All vertebrates have backbones. Therefore, all mammals have backbones.
- (B) Some dogs are pets. Some pets are cats. Therefore, some dogs are cats.
- (C) All cars have wheels. Bicycles have wheels. Therefore, cars are bicycles.
- (D) If it rains, the grass is wet. The grass is wet. Therefore, it rained.

Answer: (A) Solution: The original argument is a valid Barbara syllogism: $A \subseteq B$ and $B \subseteq C \implies A \subseteq C$. Option (A) mirrors this transitive inclusion structure: Mammals $\subseteq$ Vertebrates $\subseteq$ Have backbones.

P4.8

[Practice | MCQ | 1 Mark]

Identify the Discrepancy:

“A software company launched a massive marketing campaign that increased website visits by 200%. However, the total number of paid software subscriptions remained unchanged.”
Which of the following best explains this discrepancy?

- (A) The marketing campaign attracted broad traffic seeking free trial tools rather than users with purchasing intent.
- (B) The software code was rewritten in Python.
- (C) The company hired three new marketing executives.
- (D) The company’s competitors reduced their advertising budget.

Answer: (A) Solution: If the new visitors were looking only for free trials, traffic would surge while paid subscriptions remain flat, explaining the discrepancy.

P4.9

[Practice | MCQ | 1 Mark]

Reading Comprehension:

“Ecosystem resilience depends not merely on species richness, but on functional redundancy—the presence of multiple species that perform similar ecological roles. When disease or climate shocks diminish one species, functionally redundant species sustain nutrient cycles and energy flows, preventing trophic collapse.”

According to the passage, functional redundancy is critical because:

- (A) It allows alternative species to maintain essential ecological roles when shocks impact a particular species.
- (B) It maximizes competition and eliminates weak species.
- (C) It reduces the total number of species in an ecosystem.
- (D) It guarantees that climate shocks will never occur.

Answer: (A) Solution: The passage states that when a species declines, functionally redundant species step in to sustain nutrient cycles and energy flows, matching (A).

P4.10

[Practice | MCQ | 1 Mark]

Identify the Logical Flaw:

"Senator: Either we approve the proposed budget in its entirety today, or the nation's economy will collapse by next week."

The Senator's argument commits which fallacy?

- (A) False Dilemma (Bifurcation)
- (B) Straw Man
- (C) Slippery Slope
- (D) Post hoc ergo propter hoc

Answer: (A) **Solution:** The Senator presents two extreme alternatives as the only choices, ignoring intermediate compromises like amending the budget or passing a continuing resolution (False Dilemma).